I only ask you to submit Part C, which I believe to be representative of how you will apply these concepts in your career. I strongly encourage you to attempt all of the problems, since exam questions will be similar style to parts A and B due to the time limit on exams. Please submit your solution in PDF format on **Gradescope** by **noon on Wednesday, September 23. You must type or typeset your responses.** You must select on Gradescope which page of your upload corresponds to which numbered problem. Do not include the text of the question in your submission. **I do NOT accept late submissions** because I will post the solutions on Blackboard at the deadline.

I expect Part C to take several hours without help from AI, but with a calculator. You can use AI, but I strongly encourage you to not use AI for your first attempt, and you will lose points if I determine that you asked AI to solve everything without any effort from you. You must show at least the main steps of your work, but you do not need to show every line of arithmetic.

Quotation convention. Exchange rates are written BASE/QUOTE. The base currency is always 1 unit, and the rate is the price of that 1 unit in the quote currency. For example, USD/HKD = 7.8000 means 1 USD = 7.8000 HKD. To convert base into quote, multiply by the rate; to convert quote into base, divide by the rate. Assume no taxes.

Part A: Multiple-answer multiple choice

There may be one or several correct choices. For similar questions on exams, you must select all *the correct choices and none of the incorrect choices to receive full points.*

1. **Quotations and cross rates**
   At the start of 2020 the spot rates were USD/HKD = 7.7500 and USD/JPY = 108.00. At the start of 2026 they were USD/HKD = 7.8000 and USD/JPY = 150.00. Ignore bid-ask spreads. Comparing 2026 to 2020, which statements are correct?

   a) The USD appreciated against the JPY.
   b) The USD depreciated against the JPY.
   c) The HKD appreciated against the JPY.
   d) The HKD depreciated against the USD.
   e) The JPY appreciated against the HKD.

2. **Order books and liquidity**
   Which statements about a limit order book are correct?

   a) A market order to sell a quantity larger than the size posted at the best bid is filled entirely at the best bid.
   b) The bid-ask spread compensates a market-maker for inventory risk and for the risk of trading against a better-informed counterparty.

c) A limit order to buy at a price above the current best bid but below the current best ask becomes the new best bid.

d) For an order of a given size, slippage is larger in a more liquid market.

e) The tick size is the smallest price increment allowed in the market.

3. **Triangular arbitrage**
A dealer quotes the following executable spot rates, and you may ignore bid-ask spreads: GBP/USD = 1.2500, USD/HKD = 7.8000, and GBP/HKD = 9.8500. Which statements are correct?

   a) The implied cross rate is GBP/HKD = 9.7500, so GBP is overpriced in the direct GBP/HKD market.

   b) Starting with HKD, the profitable cycle is HKD $\rightarrow$ USD $\rightarrow$ GBP $\rightarrow$ HKD.

   c) Starting with HKD, the profitable cycle is HKD $\rightarrow$ GBP $\rightarrow$ USD $\rightarrow$ HKD.

   d) If a profitable cycle exists for a trader who starts by holding HKD, then a profitable cycle also exists for a trader who starts by holding USD.

   e) If the deviation from the implied cross rate is large enough, both directions around the triangle are profitable at the same time.

4. **Covered interest parity**
The spot rate is USD/HKD = 7.8000. One-year interest rates are $i_{\text{HKD}} = 4.0\%$ and $i_{\text{USD}} = 5.5\%$. There are no transaction costs, no capital controls, and no credit risk. Treat HKD as the home currency. Which statements are correct?

   a) CIP implies a one-year forward rate of USD/HKD $= 7.8000 \times \dfrac{1.040}{1.055}$.

   b) CIP implies a one-year forward rate of USD/HKD $= 7.8000 \times \dfrac{1.055}{1.040}$.

   c) The USD trades at a forward discount against the HKD.

   d) A Hong Kong investor is indifferent between a HKD deposit and a USD deposit whose proceeds are sold forward at the CIP forward rate.

   e) CIP is an equilibrium condition that holds only if investors are risk neutral.

5. **Uncovered interest parity**
Which statements about UIP are correct?

   a) UIP is enforced by the same riskless arbitrage trade that enforces CIP.

   b) Combining CIP and UIP gives the unbiasedness hypothesis $F_{t,T} = \mathbb{E}_t[S_{t+T}]$.

   c) UIP predicts a slope coefficient of 1 in the Fama regression, but estimated slopes at short horizons are often below zero for major currencies.

   d) The forward discount puzzle is the finding that high-interest-rate currencies depreciate by more than UIP predicts.

e) A time-varying risk premium can explain a failure of UIP even when investors form expectations rationally.

6. **Purchasing power parity and the real exchange rate**
The spot rate is GBP/USD = 1.2500. The US price index is $P_{\mathrm{US}} = 130$ and the UK price index is $P_{\mathrm{UK}} = 100$, both measured on a common base. Treat the US as home and the UK as foreign, and define the real exchange rate as

$$Q \;=\; (\text{USD per GBP}) \times \frac{P_{\mathrm{UK}}}{P_{\mathrm{US}}}.$$

Which statements are correct?

a) $Q \approx 0.962$, and the USD is overvalued relative to absolute PPP.

b) $Q \approx 0.962$, and the USD is undervalued relative to absolute PPP.

c) If US inflation is 4% and UK inflation is 2% over the next year while Q stays constant, then the USD depreciates by roughly 2% against the GBP.

d) The Balassa-Samuelson effect predicts that a country with higher productivity in tradeable goods has a lower overall price level.

e) Absolute PPP fails in part because a large share of consumption is in services that cannot be shipped across borders.

Part B: Short answer

For similar questions on exams, our answer to each should not exceed 3 sentences or 60 words.

1. A dealer quotes a wider bid-ask spread in USD/IDR than in USD/JPY. Give two distinct reasons, of the kind discussed in lecture, for the wider spread.

2. Suppose the one-year forward rate for AUD/HKD is above the spot rate, so the AUD is at a forward premium. Explain why buying AUD spot and selling AUD one year forward is not an obvious profit.

3. Covered interest parity held very tightly before 2008, but measurable deviations have persisted since then. Give one reason, and say why that reason does not simply disappear when a bank sees the deviation.

4. CIP and UIP look like the same equation with the forward rate replaced by the expected future spot rate. Explain why CIP is a no-arbitrage condition but UIP is not.

5. Hong Kong pegs the HKD to the USD and allows free capital mobility. What does this imply about short-term HKD interest rates relative to short-term USD interest rates, and why?

Part C: Order books and covered interest arbitrage

Below are the first five levels of hypothetical order books for the spot and one-year forward markets in AUD/HKD. Prices are in HKD per 1 AUD, and sizes are in AUD.

Spot				1-Year Forward			
Bids		**Asks**		**Bids**		**Asks**	
Price	**Size**	**Price**	**Size**	**Price**	**Size**	**Price**	**Size**
5.100	2,000	5.110	1,200	5.280	1,000	5.295	1,000
5.095	2,000	5.115	1,000	5.275	1,500	5.300	1,200
5.090	2,500	5.120	1,500	5.270	2,000	5.305	1,500
5.085	2,000	5.125	2,000	5.265	2,500	5.310	2,000
5.080	3,000	5.130	3,000	5.260	2,000	5.315	2,000

The one-year deposit rates are 4.0% for HKD and 1.0% for AUD. Sizes in the forward book are the amounts of AUD to be *delivered* in one year.

For parts 3) through 6), you must show your work, or at least explain your steps succinctly in words. You will not receive full credit for a number with no work behind it.

1) What appears to be the tick size in these markets?

2) Which of the two markets is more liquid? Give the two features of the order books that support your answer.

3) Before doing any arithmetic with the order books, use the deposit rates and the mid prices to work out which side of the forward market is mispriced. Then state the arbitrage in words: which currency do you borrow, and in which market do you buy and sell? Do not compute the profit yet.

- *Hint: compute the CIP-implied one-year forward rate from the spot mid price, and compare it to the forward mid price.*

4) Assume you begin with no money, and you may borrow as much as you want in either currency at that currency's deposit rate. Assume you may only trade against the best bid and the best ask in each market, and you cannot go deeper into the book. Compute the arbitrage profit in HKD, rounded to the nearest tenth.

- *Hint: one AUD bought spot today becomes* 1.01 *AUD to deliver in one year, so the spot leg and the forward leg do not have the same size. Which leg binds first?*

5) Continue to assume you begin with no money and may borrow as much as you want at the deposit rates. Now you may walk down the order books. Compute the maximal arbitrage profit in HKD, rounded to the nearest tenth, and state how many AUD you buy in the spot market.

- *Hint: start from the top level of each leg, use up the smaller size, and step down that leg. Check whether there is still a positive profit at the new, worse price. Stop when the marginal profit turns negative, then sum the profits across the levels you used.*

6) Now assume that banks face a funding constraint, so the rate at which you *borrow* any currency is 1 percentage point above that currency's deposit rate. Deposit rates are unchanged. Is there still an arbitrage at the best quotes? Explain in at most 3 sentences, and report the largest borrowing spread, in percentage points, at which the trade in part 4) would still break even.